IGCSE 0607 Practice Questions — E3.5

Asymptotes

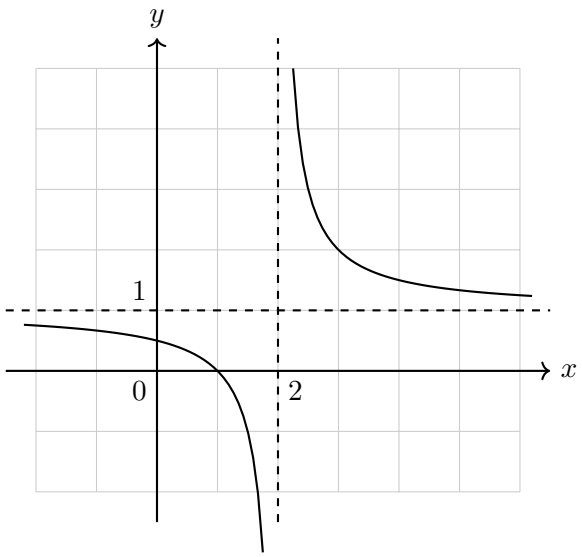
Total Questions: 18 — **Total Marks:** 101 Marks

Table of Contents & Quick Navigation

- **E3.5** Asymptotes
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

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1 The diagram shows the sketch of the graph of a function $y = f(x)$.
The dashed lines are the asymptotes to the curve.



(a) Write down the equation of the vertical asymptote.

..... **[1]**

(b) Write down the equation of the horizontal asymptote.

..... **[1]**

2 Let $f(x) = \dfrac{7}{x-4} + 3$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

3 The function f is given by $f(x) = \dfrac{5}{x+3} - 2$.

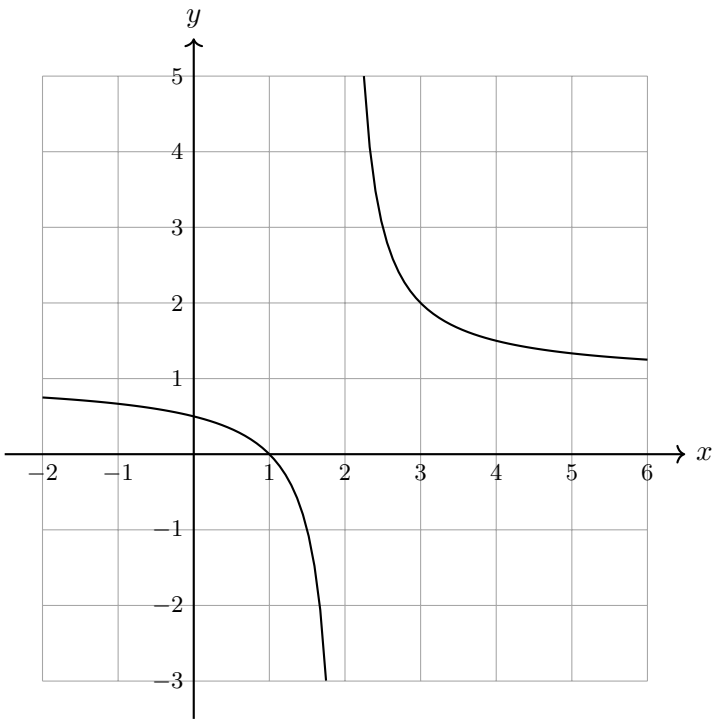
(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

$x =$ ..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

$y =$ ..... **[1]**

4 The diagram shows the graph of a function $y = f(x)$.
The graph has two asymptotes, each parallel to one of the coordinate axes.
The equations of the asymptotes involve only integer values.



(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

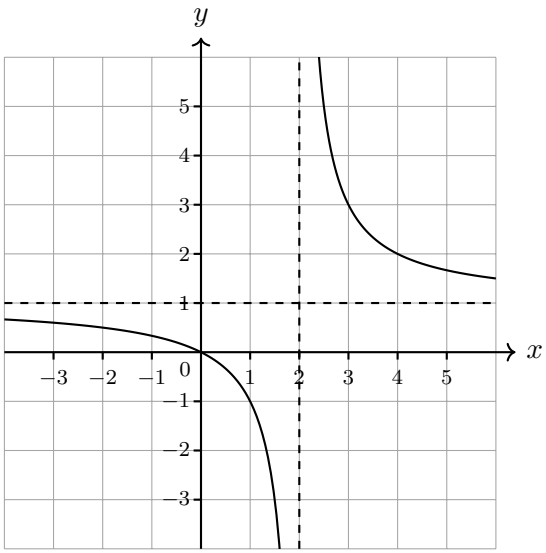
(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) The graph of $y = f(x)$ is reflected in the y-axis to form the graph of $y = g(x)$.
Write down the equation of the vertical asymptote of the graph of $y = g(x)$.

..... **[1]**

5 The diagram shows the graph of a rational function $y = f(x)$.



(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

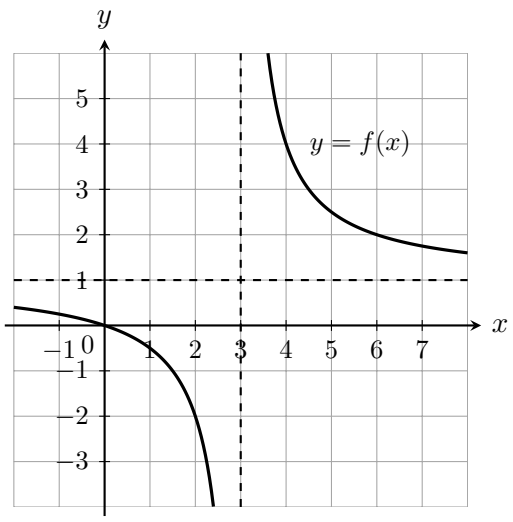
(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) Use the graph to find the value of $f(0)$ and the solution to the equation $f(x) = 0$.

$f(0) =$ ..... , $x =$ ..... **[2]**

6 The diagram shows the graph of $y = f(x)$ and its two asymptotes, which are shown as dashed lines.



(a) Write down the equation of the vertical asymptote of the graph.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph.

..... **[1]**

(c) Use the graph to write down

(i) the value of $f(4)$,

$f(4) =$ ..... **[1]**

(ii) the value of x for which $f(x) = -2$.

$x =$ ..... **[1]**

7 A function is defined as $f(x) = \dfrac{4}{x-1} + 3$.

(a) Write down the equation of the

(i) vertical asymptote,

..... **[1]**

(ii) horizontal asymptote.

..... **[1]**

(b) Find the coordinates of the point where the graph of $y = f(x)$ crosses the x-axis.

((..... , ...)) **[2]**

(c) Find the coordinates of the point where the graph of $y = f(x)$ crosses the y-axis.

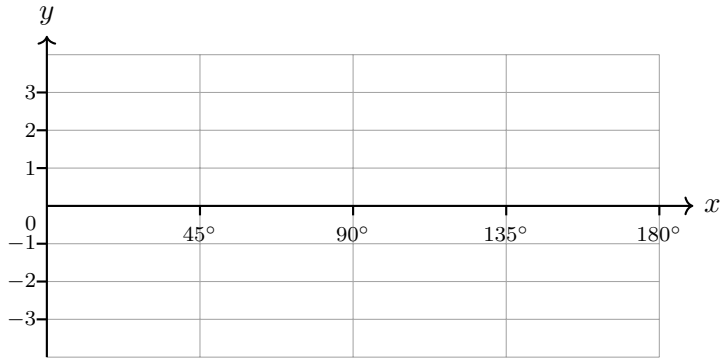
((..... , ...)) **[1]**

8 Consider the trigonometric function $g(x) = \tan(2x)$ for $0° \leq x \leq 180°$.

(a) Write down the equations of the two vertical asymptotes of the graph of $y = g(x)$ for $0° \leq x \leq 180°$.

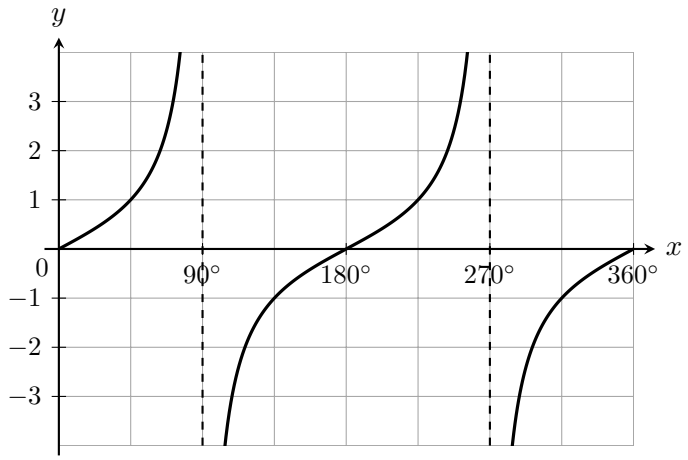
Asymptote 1: ..... , Asymptote 2: ..... **[2]**

(b) On the grid, sketch the graph of $y = \tan(2x)$ for $0° \leq x \leq 180°$, drawing the asymptotes as dashed lines.



[3]

9 The diagram shows the graph of $f(x) = \tan x$ for $0° \leq x \leq 360°$.



(a) Write down the equations of the two asymptotes of the graph of $f(x) = \tan x$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

(b) Explain how the graph shows that $\tan 90°$ is undefined.

..... **[1]**

(c) The graph of $y = \tan x$ is translated by the vector $\begin{pmatrix} 35° \\ 0 \end{pmatrix}$.
Write down the equations of the two asymptotes of the translated graph for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

(d) Given that $\tan 45° = 1$, use the graph to write down the other solution to $\tan x = 1$ for $0° \leq x \leq 360°$.

$x =$ ..... ° **[1]**

10 Consider the functions $g(x) = \tan x$ and $h(x) = \dfrac{4x-1}{2x+5}$.

(a) Write down the equations of the two asymptotes of the graph of $y = g(x)$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

(b) Write down the equations of the vertical and horizontal asymptotes of the graph of $y = h(x)$.

Vertical asymptote $x =$ ..... , Horizontal asymptote $y =$ ..... **[2]**

(c) Solve the equation

$h(x) = 0$

$x =$ ..... **[2]**

11 A function is defined as $f(x) = \dfrac{3}{x+2} + 1$.

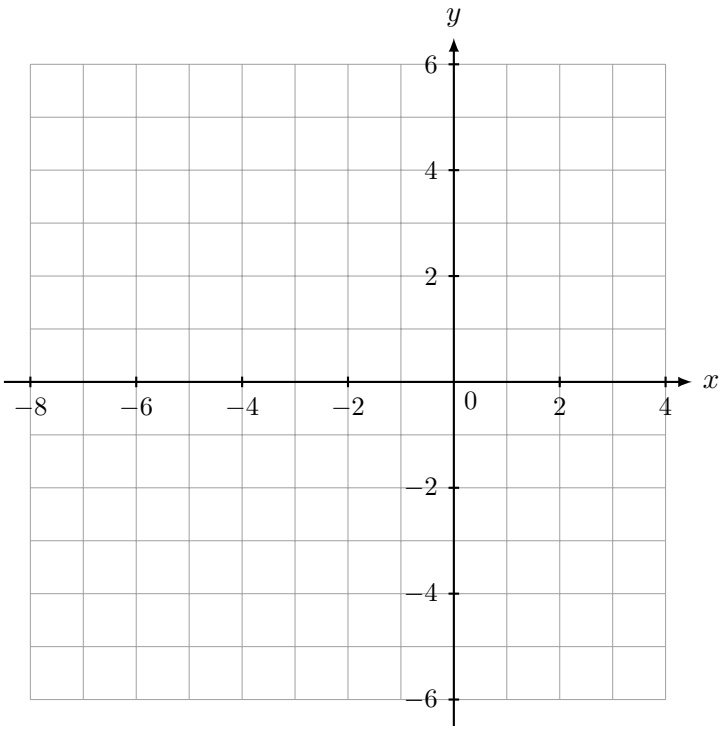
(a) Write down the equations of the two asymptotes of the graph of $y = f(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

(b) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x-axis and the y-axis.

x-axis at ..... , y-axis at ..... **[2]**

(c) On the grid, sketch the graph of $y = f(x)$.



[3]

12 Let $f(x) = \dfrac{3}{x+2} - 4$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) Find the coordinates of the points where the graph of $y = f(x)$ intersects the coordinate axes.

x-intercept = ... , y-intercept = ... **[5]**

(d) The graph of $y = \tan x$ has an infinite number of vertical asymptotes.
Write down the equations of the two asymptotes in the interval $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

13 A function is given by $f(x) = \dfrac{a}{x-b} + c$, where a, b and c are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = -1$.
The graph intersects the x-axis at the point $(5, 0)$.

(a) Write down the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[2]**

(b) Work out the value of a.

$a =$ ..... **[2]**

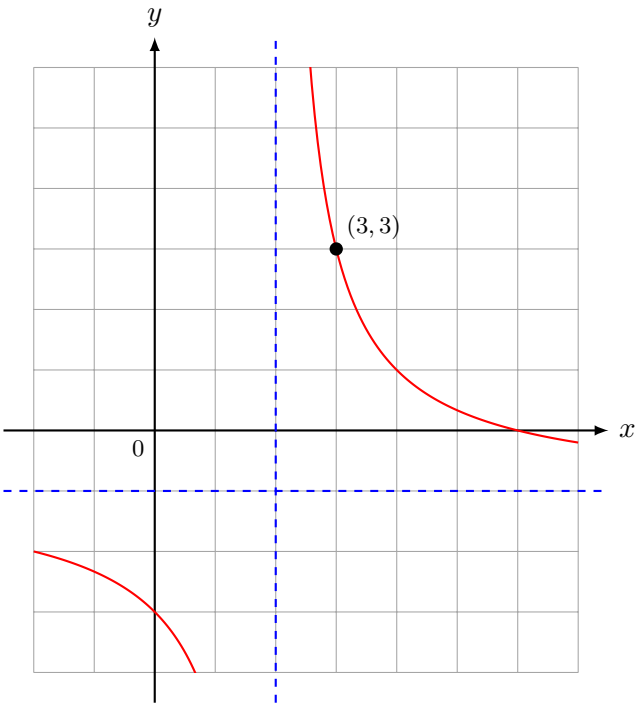
(c) The graph of $y = f(x)$ intersects the straight line $y = -x + 5$ at two points.
Find the x-coordinates of these two points.

$x =$ ..... , $x =$ ..... **[2]**

14 The diagram shows the graph of $y = f(x)$ where $f(x) = \dfrac{p}{x - q} + r$.

The graph has asymptotes at $x = 2$ and $y = -1$.

The graph passes through the point $(3, 3)$.



(a) Find the values of q and r.

$q =$ ..... , $r =$ ..... **[2]**

(b) Find the value of p.

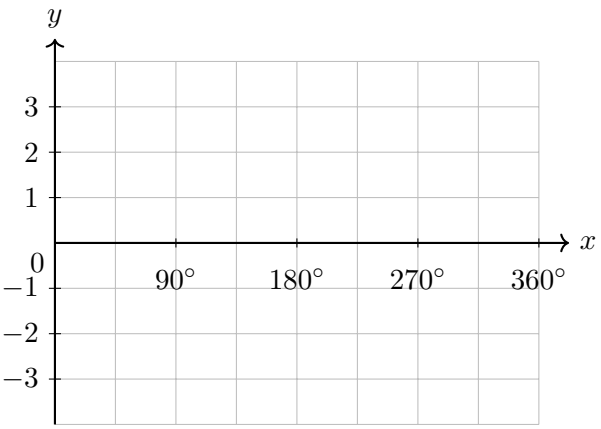
$p =$ ..... **[2]**

(c) Find the coordinates of the point where the graph of $y = f(x)$ crosses the x-axis.

(..... , ...) **[3]**

15 The function $g(x) = \tan x$ is defined for $0° \leq x \leq 360°$.

(a) On the grid, sketch the graph of $y = g(x)$.



[2]

(b) Write down the equations of the asymptotes of the graph of $y = g(x)$ in this domain.

[2]

(c) Solve the equation $2\tan x + 3 = 0$ for $0° \leq x \leq 360°$.
Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

16 Consider the function $h(x) = \dfrac{2x^2 - 5x - 3}{x - 2}$ for $-4 \leq x \leq 6$, $xe2$.

(a) Use your graphic display calculator to find the zeros of $h(x)$.

$x =$ ..... , $x =$ ..... **[2]**

(b) Write down the equation of the vertical asymptote of the graph of $y = h(x)$.

..... **[1]**

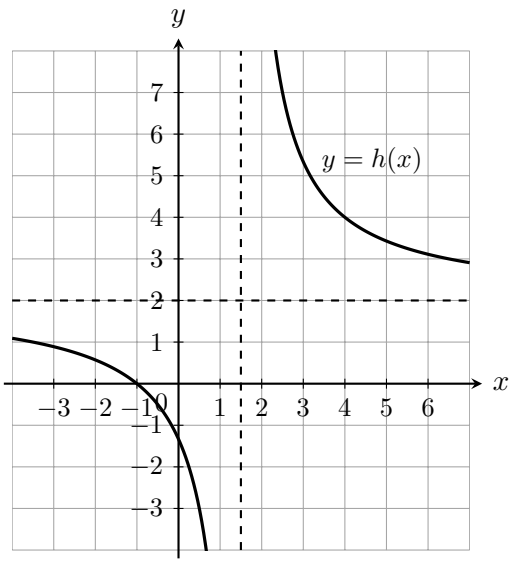
(c) Explain why the graph of $y = h(x)$ does not have a horizontal asymptote.

..... **[2]**

(d) The equation $h(x) = k$ has no real solutions.
Find the range of possible values of k, giving your boundary values correct to 2 decimal places.

..... **[2]**

17 The diagram shows the graph of $y = h(x)$, where $h(x) = \dfrac{5}{x - 1.5} + 2$ for $x \neq 1.5$.
The dashed lines show the asymptotes of the curve.



(a) From the graph, write down the equation of

(i) the vertical asymptote,

..... **[1]**

(ii) the horizontal asymptote.

..... **[1]**

(b) Explain why the equation $h(x) = 2$ has no solution.

..... **[1]**

(c) The graph of $y = h(x)$ is reflected in the y-axis.
Write down the equations of the vertical and horizontal asymptotes of the reflected graph.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

(d) Solve the equation $h(x) = 3.4$, giving your answer correct to 3 significant figures.

$x =$ ..... **[3]**

18 The function $f(x)$ is given by $f(x) = \dfrac{p}{x-q} + r$, where p, q, and r are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 5$ and a horizontal asymptote at $y = -3$.
The graph passes through the point $(7, -2)$.

(a) Write down the value of q and the value of r.

$q =$ ..... , $r =$ ..... **[2]**

(b) Find the value of p.

$p =$ ..... **[3]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(d) Write down the equations of the asymptotes of the graph of $y = f^{-1}(x)$.

$x =$ ..... , $y =$ ..... **[2]**